YATHARTH GOYAL

APPLIED AI / COMPUTER VISION / DATA SYSTEMS / OPTIMIZATION

Jaipur, India · yatharthgoyal657@gmail.com · **GitHub** · **LinkedIn** · **LeetCode**

TECHNICAL SKILLS

Languages Java, Python, JavaScript, SQL

AI & Optimization YOLO, scikit-learn, model evaluation, computer vision, SciPy / HiGHS, MILP, pandas, NumPy

Services & Data FastAPI, Express.js, PostgreSQL / PostGIS, Redis / BullMQ, REST APIs

Frontend & Quality React, Vite, Docker, GitHub Actions, Git, pytest, Playwright

SELECTED ENGINEERING PROJECTS

RouteGuard 2.0 | Applied Vision & Geospatial Research · IN PROGRESS

YOLO · Python · PostGIS · React / Leaflet · TomTom

- Developing a road-hazard detection and route-intelligence pipeline combining geospatial routing, driver workflows and YOLO-based road-damage research.
- Audited label integrity and validation splits; compared detection checkpoints. Research target is ≥95% precision and recall per severe hazard, not achieved or production validated.

AegisED | Human-Reviewed ML Decision Support

REPO · LIVE DEMO

Python · FastAPI · scikit-learn · Express · PostgreSQL · React

- Integrated human-reviewed triage-support recommendations into operational APIs with role-based controls, recorded overrides and transactional resource allocation.
- Selected a model from four candidates; documented macro-F1 0.7893 across 900 held-out synthetic test cases (not clinical validation). Source includes model and API test evidence.

EvalForge | Evidence-Grounded Assessment Engine · PRIVATE · RELEASE PENDING

Python / FastAPI · Redis / BullMQ · PostgreSQL · React · Docker

- Built rubric-grounded written evaluation with exact criterion evidence spans, instructor review and appeals, plus durable asynchronous jobs and retry handling.
- Validated criterion evidence and local evaluator behavior (7/7 tests); backend tests 15/15. Hosted deployment and full real-service browser E2E remain pending.

Grid Flex | Applied Optimization & Simulation

SOURCE

Python · SciPy / HiGHS · MILP · JavaScript · AWS SAM

- Implemented a mixed-integer energy scheduling engine for synthetic solar, grid and battery constraints, comparing cost/carbon trade-offs and grid-outage scenarios.
- Added independently verified solution constraints and solar/battery sensitivity studies; 52/52 local regression tests passed. Cloud templates are not deployed.

JAVA PROBLEM SOLVING 16-topic DSA repository with 161 catalogued LeetCode implementations (repository coverage, not verified accepted submissions). **VIEW CODE**

EDUCATION

B.Tech in Information Technology | JECRC, Jaipur — **Expected 2028**